GURDHARAM JEET SINGH

Full-Stack Software Engineer • Web Systems & Spatial Frontend Architect

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TECHNICAL SKILLS

Languages & Core: TypeScript, JavaScript (ES6+), Python, Go, SQL, HTML5, CSS3

Frontend & Spatial Web: React 19, Next.js 15 (App Router, Server Components), Tailwind CSS 4, Vite 6, Framer Motion, Lenis Smooth Scroll, Three.js, WebGL2 (Camera Kit), Core Web Vitals (LCP, CLS < 0.02)

Backend & Distributed Systems: Node.js, Express, FastAPI, PostgreSQL, Supabase, MariaDB/SQLite Failover, Redis & BullMQ, WebSockets, REST APIs, Docker, Linux/Bash

Cloud & Platform Infrastructure: Official Meta WhatsApp Cloud API (WABA), GitHub Actions 24/7 CI/CD Fleets, Railway, AWS (S3, Lambda), Exotel Telephony, Gemini AI

KEY PRODUCTION PLATFORMS & SYSTEMS

SOLUM Minerals — Procedural 3D WebGL Bio-Carbon Platform **2026 (Oct) – Present**
Lead Frontend Architect • Next.js 15, Three.js, WebGL2, Tailwind CSS Open Source

- **Procedural 3D WebGL Rendering:** Engineered code-generated Three.js WebGL tablet model synthesizing 1024x1024 noise textures procedurally in memory via Canvas API, eliminating 15MB+ of 3D asset downloads.
- **Scroll Kinematics & Core Web Vitals:** Implemented scroll-linked camera interpolation and river kinematics in Next.js 15 App Router, achieving a 99+ Lighthouse score with zero layout shift (CLS < 0.02) and sub-second FCP.

Dream Heights — Luxury Spatial Real Estate Portal **2025 – Present**
Spatial & Frontend Architect • React 18, TypeScript, Tailwind CSS, Framer Motion, Lenis Open Source

- **Spatial Architectural Pins & Navigation:** Built luxury residence exploration portal in React 18 and Vite 6 featuring spatial residence pins, dynamic floor-plan modals, and custom interactive architectural emblems.
- **Hardware-Accelerated Physics & Inertial Momentum:** Integrated Lenis 1.1 smooth scrolling with Framer Motion layout transitions, maintaining silky 60 FPS rendering across high-DPI viewports.

EduCore — Enterprise College Management ERP **2025 – Present**
Lead Full-Stack Architect • React 19, TypeScript, Express, PostgreSQL, MariaDB, Railway Open Source

- **Dual-Engine Relational Architecture:** Engineered multi-tenant campus ERP supporting PostgreSQL and MariaDB connection pools with automatic connection health recovery, parameterized query sanitization, and sub-15ms latency.
- **9-Role RBAC & Dynamic Schema Builder:** Built strict JWT role-based access control across 9 permission tiers, paired with a runtime form schema builder and forensic audit trail logging actor IP, device profile, and data diffs.

Enterprise WhatsApp AI & Lead Management CRM **2025 – Present**
Lead Architect • Meta Cloud API, Node.js, Express, Supabase (PostgreSQL) *Production Platform*

- **Official Meta Cloud API Architecture:** Engineered multi-tenant CRM gateway integrating Meta Graph Cloud API (WABA ID: 106690...), handling bi-directional messaging, template dispatch, and HMAC-verified webhooks.
- **24-Hour Autonomous Follow-Up Radar:** Built real-time engagement engine with dynamic cadences (24h/36h/48h/72h), tracking lead conversation states and triggering automated follow-ups via Supabase.

WORK EXPERIENCE

Gurdharam Software Consultancy • Systems & Web Engineer **2025 – Present**
Independent Software Consultancy • Remote *Punjab, India*

- **Telephonic Voice AI & High-Throughput Queues:** Built real-time telephony agent over Exotel WebSockets, SIP, and VoIP, achieving <700ms turn latency across Sarvam AI Indic LLMs, Groq LPU, and Silero VAD, with Redis/BullMQ processing 25,000+ daily events.
- **Branders Digital Agency Platform:** Architected client agency platform featuring interactive Three.js 3D viewport, CSS3 physics navigation, and mobile-hardened layouts designed for high-conversion client acquisition.

EDUCATION & CREDENTIALS

Baba Farid Group of Institutions (BFGI) — B.Sc. in Computer Science **Expected May 2027**
Core: Data Structures & Algorithms, Database Systems, Computer Networks, Operating Systems *Punjab, India*

Credentials & Verification: Meta Verified Business Entity • Official WhatsApp Business Platform Developer • Sarvam AI Startup Partner • Govt. MSME Registered Enterprise (UDYAM-PB-03-0079744).